SECTION 11 24 24

PERMANENT HORIZONTAL FALL PROTECTION, METAL ROOFING
08/20

PART 1 GENERAL

1.1 DESCRIPTION

Standing Seam Metal Roofing (SSMR) fall protection to be used in all SSMRs with slopes greater than 3/12.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

UNITED FACILITIES GUIDE SPECIFICATIONS (UFGS)

07 61 15.00 20	Aluminum Standing Seam Roofing

UNIFIED FACILITIES CRITERIA (UFC)

UFC 3-301-01	Stuctural Engineering: Fall Prevention and Protection

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A10.32	Personal Fall Protection Used in Construction and Demolition Operations
ANSI Z359.1	Safety Requirements for Personal Fall Arrest Systems, Subsystems and Components

OCCUPATIONAL SAFETY AND HEALTH ORGANIZATION (OSHA)

1926.502	Fall Prevention Systems and Criteria and Practices

1.3 DEFINITIONS

Roof sloping (#/12) refers to the number of inches a roof rises in height for every 12 inches, as measured horizontally from the edge of the roof to the centerline.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

For fabrication showing the complete fall protection system. Layout drawings of each system in relation to the supporting structure indicating the locations of properly labeled components.

SD-03 Product Data

Manufacturer's data and product information indicating the sizes, descriptions, capacities, test certifications, and other descriptive data showing in sufficient detail that the product complies with the contract requirements.

SD-07 Certificates

Installer's Certification

SD-08 Manufacturer's Instructions

Maintenance Procedures: Including parts list and maintenance requirements for all equipment.

Operation Procedures: Indicating proper use of equipment for safe operation of the systems.

1.5 QUALITY CONTROL

After fall protection system is installed, the manufacturer's authorized representative shall inspect and operate the system and make any final adjustments. The manufacturer's authorized representative shall issue a certificate attesting to the system's design and installation, and formally submit to the USG

1.6 QUALIFICATIONS

a. Furnish proof of installer's certification approval by manufacturer in the form of the installer's current certificate issued by the manufacturer.

b. Provide a designer's qualification statement.

1.7 WARRANTY

a. Provide lifetime manufacturer warranty

b. Submit manufacturer warranty and ensure that forms have been completed in Owner's name and registered with manufacturer

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Design, furnish, install, and certify a complete and useable permanent fall protection system, inclusive of all components, appurtenances, fasteners, end stops, carriages, and equipment necessary for safety of maintenance workers.

Submit horizontal life line and anchorage connector systems for approval by the designer of record and the installation public works group's cognizant architect or designated fall protection expert. The chosen fall protection system must be compatible with the Base Exterior Architectural Plan, low-maintenance, designed for service in harsh coastal conditions, and compliant with ANSI/ASSP Z359 requirements. The following proprietary systems would meet government requirements, but other options may be proposed by the contractor, subject to final approval by the government:

1) 3M Capital Safety Roofsafe Rail System
2) Miller ShockFusion Horizontal Lifeline Roof System
3) Kee Safety KeeLine Engineered Lifeline Solution

2.1.1 Design Requirements

a. Design must be performed, signed, and sealed by a Professional Engineer from the manufacturer experienced in the design of horizontal lifeline fall arrest systems. Submit qualifications of engineer.

b. Design must provide for access from the roof hatch (where provided) to the roof ridge, allowing for continuous worker tie off to reach the ridge line. For buildings without a roof hatch, design must provide for access from the roof eave (at an acceptable location) to the roof ridge, allowing continous worker tie off to reach the ridge line. D-rings are acceptable for ascent and decent parallel to standing seams, or propose alternative method.

c. Low profile and architecturally pleasing systems are preferred, and will be evaluated higher than more obtrusive designs.

d. Concealed fastners are preferred.

2.1.2 Performance Requirements

a. System must be able to accomodate three (3) users at a time.

b. All fall protection anchorages must be designed for a minimum 3,100 lb nominal live load capacity. Anchorages, including any supporting structure, must be designed to allow future replacement of the horizontal lifeline system or anchorage connectors regardless of the required capacity of the chosen horizontal lifeline system or anchorage connectors. Anchorages that support two or more independent horizontal life line spans at a single point must be designed for the combined load effects of the horizontal life line systems in accordance with ANSI/ASSP Z359.6.

2.2 FINISHES

a. Choice of finishes in powder coated or anodized aluminum alloy are preferred.

b. Color must comply with the MCBCL Base Exterior Architectural Plan (BEAP) and the USG Architect. Submit color samples for approval.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

After fall protection system is installed, the manufacturer's authorized representative shall inspect and operate the system and make any final adjustments. The manufacturer's authorized representative shall issue a certificate attesting to the system's design and installation, and formally submit to the USG.

PART 3 EXECUTION

3.1 INSTALLATION

Fall protection system shall be installed under the direction of manufacturer's authorized trained personnel.

3.2 INSPECTION

The manufacturer's authorized representative shall inspect and operate the system and make any final adjustments.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Demonstration

The manufacturer's authorized representative shall issue a certificate attesting to the system's design and installation, and formally submit to the USG.

3.3.2 Training

Train designated US Government personnel on-site regarding proper use of system.

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